

Shagor Rahman: “Humanity’s Story: A Search for Moral Clarity”

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the next recorded talk is 50 minutes long By shager Ramen Humanity story a search for moral Clarity hi everybody so I'll be going over what I'm calling Humanity's story a search for moral Clarity and I'll get into kind of what that cheeky title means um going over kind of myth morality narrative self- understanding and um the meaning of life and no that one was too far but I guess not too much farther than one I'm claiming um so yeah before we get started I just want to give a big thank you to the active inference Institute um you know for overall the work they do um it's been super rewarding um and engaging to to be a part of uh I put my information here so feel free to reach out if anything picks your interest or you know bothers you or you have to correct anything feel free to um give me a shout so we'll be discussing um you know a few things as I mentioned so one of the big roles um that I feel active inference can um you know one of the ways that I can at least intuitively scale into human culture now first of all through our use of symbols kind of what makes us unique as a species um can be understood semiotically or through symbols um and the road there uh as I make the case in myth of objectivity is through morality um largely through bodifying norms and expectations which again ties very nicely into Act of inference I believe the last way is to narrative and specifically I'll be talking about uh kind of at a meta level The Narrative of modern narrative and and you know why are we why are we doing this you know it's you know epistemic reasons interesting as it is but also I think that kind of applying a r rigorous framework to humanity specifically particularly as we get into narrative um we'll see it more broadly tackle um some other areas of humanity um this is necessary I feel for cultural cohesion um something generally lacking these days um you know with partisan um and in polarizing um the nature of our culture at the moment and in our politics uh and so it'll achieve hopefully or at least understand the outlines for cultural cohesion uh and things like social and personal uh integration and I'll do this by as I mentioned by expanding upon my myth of objectivity hypothesis right and so my own road to to um becoming interested in the free energy principle

and active inference was being disillusioned with the economics Paradigm um finding it to be as as I'm sure other folks were um insufficient to describing human behavior even through the kind of quote unquote narrow lens of the marketplace conceiving people as a neoclassical framework does as utility maximizing um individual agents who are well-versed in what their own preferences and interests are um you know it's kind of quite unrealistic to say the least not only that but behavioral economics didn't really um do it for me either um it also held up this idea this neoclassical idea of rationality um as the kind of goal standard and you can see that with some of the recommendations you know the way we should um you know default constantly default a 401ks um you know maybe that makes sense but it does it comes up with that advice it justifies the device using this kind of principle of stationarity meaning an agent's um preferences should not change over their lifetime which of course is not not realistic and so of speaking of um know behavioral paradigms that are lacking um you know at first I thought well the neoclassical model on its own isn't interesting um maybe we can marry that with you know some views of moral philosophy because I'm sure moral philosophers have you know a much better view of of humanity that's what I thought uh so here are a few uh few different moral philosophies that maybe some of you folks are aware of maybe others not um but kind of briefly um deontology utilitarianism are you know traditional from the enlightenment era going back to Kant um and Stuart Mill um paradigms of moral philosophy um interpretivism is a bit more recent but um I think still been around for for quite uh some time um and the last one is uh just a couple examples of some Eastern philosophy and so when we think about I guess the question I'll pose to you know those listening is what type of Thermo Dynamic systems in terms of um their adaptability um you know how do we characterize these various different systems you know are they single equilibria are they steady States um are they uh maybe chaotic maybe they don't reach any equilibria um you know how do we think about these um so firstly deontology uh and utilitarianism um and you know if not familiar with them um don't worry I'll come explain it but um the ontology and utilitarianism um I view as single Lake RBM even though they are uh different um they seem to have one um kind of one unifying mistake which is you know the environment or kind of context doesn't change or in other words there is no uncertainty in these systems that they assume you know by their prescriptions of morality and moral philosophy so deontology assumes there are certain set of rules that you should never violate that um never change over time um so you maybe thou shall not kill th shall not steal etc etc um there's no conditions under which those should be violated um and utilitarianism this is all of your actions are based

their morality at least is um evaluated based upon the impact that they'll have on the the aggregate welfare of others right so um this is kind of featured for instance in uh the good place for those um watched it but the problem with both of these is that um you know in a word it's uncertainty right there's no epistemic foraging there's no model for how Agents come to learn okay what are the rules um maybe the deontic rules that I should follow what are the ways that my actions impact others like what is that process of um socialization that kind of awareness those you know the empathy um you know all the things that you know you know should I read a book should I read fiction um you know how should I speak to people um you know all these are essentially epistemic tasks um of learning tasks that are not featured into you know these twin systems um and and I'll get into perhaps what this reveals about Western Society um of spoiler but this is the kind of analytical bias of Western Society I think revealing itself in deontology and and utilitarianism specifically so interpretivism um this comes out of postmodernism I believe or at least it was um rebr into the dialogue following modernism into postmodernism Etc so it says essentially kind of rejecting these uh analytical reduction single equilibrium as I presented it here um you know frames of moral philosophy and says that well it's based on the agents interpretation of their behavior you know what are the context for their actions also how do they interpret their actions etc etc well of course it's an important element of morality um the kind of classification that at least I would label is chaotic right so if if we're just saying it's up to someone's interpretation well how should they interpret it um should everyone interpret it differently is everyone's interpretation equally accurate how should we you know yada yada the point is that it's not super helpful for society to um as a approach to morality all right so final two confusion is and Daoism um uh I see them as being adaptive and being steady state um focus on the kind of rituals and behaviors not content specifically not focus on particular deontics um and not kind of hyperfocused on the impacts of behaviors to others Al that that can be a component as can you know interpretation but um it's really about the emphasis of these two paradigms um kind of inherently acknowledging um that the world is changing that Society is changing and yet the goal of um morality at both personal and social level um is to find coherence so it's kind of still privileging um some kind of equilibria um some kind of Harmony but not kind of how you get there and I'll get into how it it does that um with active inference so you know active inference what I would say is that of Daoism and um Confucianism they come from a tradition that um Carl Yung would label as a the mytholog mythological construct of the cosmic human or Cosmic man um aka the symbolic self which is the you know term I'll use for kind of going forward because it kind of ties into

um you know my theme of symbolic agents Etc so the point is that uh you know Carl after serving um doing a cross-cultural analysis of various different mythologies and and traditions found that this particular what he called Arch type reemerged uh and essentially the cosmic human was the perspective that we all kind of part we all kind of connected to one another um not only to other people but to the cosmos into the world um in general and our actions do have consequences because of that inherent interconnections um moreover we do have a certain responsibility to you know the world as a whole to other people um because of our place as a kind of a node in this hierarchy this I'm signaling where I'm going with this um but it allows as you can see um an area both for the kind of characteristic uh States so the ways to minimize uncertainty kind of using both that you know epistemic and exploitative um you know Paradigm of active inference um but kind of does so specifically um by conceiving this kind of broad view of the self right and so here of course is how it connects to active inference so active inference unlike the neoclassical framework and unlike other um single equilibrium um approaches to social science and to science in general it allows um noes to communicate um both with spatially and temporally and so kind of overtime an individual um passes information to itself or to kind of quote unquote its its future self and um in some ways it's its path self in terms of goals um but the self is not a stationary construct it is something that is constantly seeking um who what other nodes are kind of included in my hierarchy um what other are including kind of myself um and these symbolic self is an example of our ability to exate all these different nested nodes so our the result of us being a symbolic agent is that every time we meet someone say you know who says they're Canadian um well if we have met this person before our model of what it means to be Canadian now includes that person we now have a you know instance of the con the general concept of a Canadian of that kind of General archetype so that means that the model we have of Canadian whether or not we are Canadian um we have the ability to compute to constantly compute what the self is um and by virtue what the other is and so this the kind of Road the tra the trajectory that I outlin that kind of brought us to here and what it means and all that um I draw from our social bias so what are kind of some instinctual impulsive elements that first um started to see Human Society expand kind of outward laterally or even over time well at least laterally is kind of um within a specific time uh I make the case that it is social bias something that we are inherently born with um you know although we are not um you know born racist per se um we are born with a specific bias and that's accent bias or shth it does seem to be something that emerges in utero and we have preference for um pronunciations for vocal accents uh over others those we are

more familiar with while that's all negative it does have clear biological benefits of course that's why it exists and that's why it you know persists today um and secondly what what do we do with a society that's based on people with the same accents um well that's an inherently un instable one when that's not um super adaptive AI is extensive and allows for greater cooperation um it's unstable because it allows for deviance free writing things like that okay so enter morality enter morals um and so here now we're connected in this um symbolic node Way by codifying those Expectations by minimizing surprise using morality right so this is my myth of objectivity um you're supposed to behave cooperatively you're supposed to follow these Norms by virtue of where you where you've grown up you know that is social bias that is shth um evidently you are not doing that so how do I deal with that surprise well over time of course that model you have um the shared model you have of um you know that's based on identifying with those like you starts to wear thin starts to break down and so in comes morality as a way to stabilize um to offer kind of a hood that you place on um deviants and that you can then you know use to shame and guilt um into aligning with the shared model um or of course there's punishment ostracization and other things right but that's what the the tool that morality probably had in the plene and continues to have with us today all right okay that third leg is um okay so if this is itself self um what are we modeling off of right so we have the internal model Act of inference we have the mark blanket that scream between us and the world so then we have the external States so what once we have kind of symbolized itself what is the world what exactly are we modeling off of sometimes it's the world you can say okay let think about um farming think about you know someone who's born and raised as a farmer the land will feature very prominently in their model of the world think about a hunter land will figure um you know the world itself the physical world will um you know will mean something um however it won't mean everything so what are some other things that will constitute the world probably the other right so social bias of course Works into that there's those who sound like you there's those who's not or those not like you um so by virtue of having a bias means that you're biased away from other people and that's why we don't like to admit that we have biases um and that's why we were troubled when um people tell us that we're biased so that's one example over the trouble um with the other and you can think about you know the Cold War think about any time that you know a country is um you know under attack the other becomes prominent becomes Salient um the in group binds together and you know for a while um you whether it's 911 or whether it's World War II there's kind of great social cohesion trouble of Wars end um enemies become as Salient um and then what happens

well this is you know how we have this is you know enter kind of Mythology not these things are entirely separable um there's a lot of overlap but um you know for our purposes if we think of you know mythology and we start to think of um narrative what we're talking about is uh in the language of Mel in fields we're explicating the reference frame we're explicating the measure the kind of yard stick of what's good what's bad um whether is you know according to Mike 11 the frame of reference that allows agents whether you know bacteria or complex organisms to judge whether they should approach something or they should avoid it is it an internalized um plane or frame of reference um what we're doing eventually once we get to myth once we get to um explicit morality once we get to narrative is we're exola that frame of reference we're saying okay um you know these group of people say are evil um you know we we're all descended from this um ancestor you know from you know Hercules from bolf from this kind of half man half god um we explicating these reference frames to gain cultural solidarity decaying cultural cohesion right and so um I mentioned this uh this is more going into my um myth of objectivity hypothesis but I'll just kind of briefly show it because you know I would love to get some feedback from the folks listening to this um so this is how I have um I hope to formalize um you know how we um you know became symbolically literate um if you will so that kind of passed from shith to morals um to you know full symbolic agent um and so I just have kind of an agent here um assuming that another Agent B has a fellow X and therefore shares the model of the world has some error you know the error reaches some threshold um and then suddenly you know if then else um you end up um with a new model it eventually leads you to um modeling the prediction error so here have um a somewhat familiar equation but kind of modified but instead of um uh taking the where the posterior is kind of conditional on um traditionally conditional on the model um here in place of that I have you know an objectified error um so the point is that you have the these categories of whereas you know typically the prediction error will just be whatever kind of noise that you experience that are not in line with the prediction um and other you'll select a different prediction model um different model at that point um or you just assume it away as um as noise in this case there is an explicit model of error so when someone decides to steal food when someone decides to um not help with uh childbearing when someone decides um I can't think of any um moral deviations um but um the point is that there is a specific set of deviations from the typical predictions versus just being um surprised generically so it's a modeled version of error um that then allows uh an agent a to view a given agent um as possessing the ability to select between adherence to a higher level or superordinate model um now an objective model um of

the group or a narrow one so they can comply or they can deviate that's a C and D um subscript and the point is that um or you know I guess one one takeaway is that okay this is just metacognition um that of upper level superordinate G um and this is transc meaning that it's not um purely objective um it's framed as objective from um the point of view of the agent but that only means that they're transcending their own subjectivity and the subjectivity of the agent um by conceiving that they have the ability to select between a higher level or a lower level model and so in this kind of simple way at least at this level of explanation we can see how from an active inference point of view we could have gotten to something like objective morality uh and more importantly something like Free Will and of course the various different significance and so just um again a couple you know I view this as um the kind of a hypothesis hiding in plain sight why because morality is something that we're so impulsive about um something we have so many emotional um ties to um you know for instance um we're constantly looking for vict um we're constantly claiming to be the victim we're constantly looking for um people who victimize um and yet we can't really agree on what the specific contents of morality should be even though as I'm when I mentioned deontology I'm sure there was folks listening who had ideas of what could be deontic meaning forever forever there um so went over the some of this um you know there's you know social biasing U morality there's impulse to do this um and there some kind of research that I've done um in the economics going back to that neoclassical Paradigm although I was kind of dumping on it earlier um what these single equilibrium analyses allow for uh is the ability to say okay where does this model break down where does this model fall so you might say it falls down everywhere that's not quite true um people are uh Salish they do have um stationary preferences at least for um a long enough time for that model to be useful um however it does fall down in certain interesting ways one interesting failure of the single equilibrium economics Paradigm is the need for social trust um and so social trust means that we have certain expectations we have expectations that company selling something sells what they promise to sell um it means that you can you know trust the market system trust the institutions um in the market that you inhabit and engage in economic activity um so you think that this at all levels you essentially need it you essentially need to take medicine and understand that hope the people are um you know so ethically inclined um you know because of course if um they do something bad they might not get caught for it y yada the point is that there needs to be some sort of um agreement on social trust so it's an example of how kind of morality seeps its way into something even as um even as intended to surgically remove all

kind of humanity from the Paradigm all all morality all essence of morality from econ um the study of Economics social trust and morality seeps back into it um the same thing with um Linguistics um won't get into this one too much now but the de of transcendental signify so it's a way of structuring otherwise of elusive um set of symbols linguistic systems um we always have something there's always as I mentioned there's always God there's always some sort of myth that's tying everything together um even in secular society we have not um you know we don't have kind of uppercase God G God the way you might might have in the past you know there's a lot of different religions there's a lot of Institutions and probably agnosticism and secularism is kind of increasingly dominant um so what do we have in this case we have democracy we have um science we have kind of other um essentially kind of mantras if you will um that require this kind of shared belief um element which kind of has very moral characteristics but is is kind of tied back into um a kind of shared belief system um that then Heathers our uh you know norms and expectations too and allows for us um they kind of have you know social and political dialogue um yeah um I SP yeah I spent a bit of time on the mythity hypothesis um quickly um I don't know if I'll have time for uh this entire run through um but once we have established that um we achieved a symbolic capacity um of what is uh the next step do we just kind of walk around shouting symbols um no we have kind of narratives that uh attached to our symbols it's the way we kind of piece them together it's how we connect one symbol to another at a very um kind of crude way of describing this um there's a recent paper on narrative Theory active inference um which I encourage um you know folks to go out there and and um and seek um it covers episodic memory it covers narrative identity as so some interesting topics throughout on narrative um the one thing that I wanted to add is um it makes the claim that narrative is for the future it's for prediction just as many things in um active inference are however I think that you could equally say that in order to establish um you know a productive prediction um we write narratives about our past so we can separate you know the person we were in high school um the person we were um you know in our 20s our 30s who we can we tell stories about that about our last job in order to draw a boundary between us um and that person so it's a way of um of inheriting in in um inhabiting various selves um but not identifying not overidentifying with the past one so I just want to add that because I don't think it was covered in the uh in that paper um and so here I have a couple different of the core elements of narrative and why I think that it it SS so well in the The Narrative Paradigm so I think the story describes as a set of you know the underlying events um so it's not explicitly what's in the narrative so if we read a novel um and you know

the the novel is a short story about um you know Harry Potter it's uh exactly what's happening in that version of Harry Potter maybe he's um you know already at school doesn't kind of go into to his backstory necessarily um um but this but the the story is all the things all the underlying events that happen that are related that are necessarily depicted in that narrative um and so this is essentially the the generative process right so it's the outside world that's um you know that's being modeled in um The Narrative and in the kind of causal chain that is the plot um so the point is that um uh kind of kind of quick but the point is that when discussing or when of analyzing um analytically the core elements of narrative um you know a lot of the themes that emerge are kind of similar to um at least some of the key elements of um active inference yeah skip this one for now um yeah so um one thing I think it's important to notice is that there is implicit versus kind of explicit um narratives um and so there's this there's the idea of of feature placing um so this is essentially how um vocal vocalizations you know emerged prior to symbolic CU so you know other animals have um verbal Expressions but they're non symbolic um a feature placement is um kind of one single verbalization um that might be some way that we can think about how um language began um um and so essentially the kind of going back okay Concepts need context if we just s fire um what does this mean does it mean that I'm fired from a job does it mean that something is burning um or does it mean that you know I'm just um you know someone said something go really cool and I'm just saying fire uh in that kind of slang context um so exping the explicit so um here uh I just wanted to raise um the instance of where creating narratives um although everything can be narrated um not everything um is narrated um so there could be unconscious events that happen in your past or childhood or even during the day that you haven't really thought about so um various different therapeutic techniques um address that so internal family systems is one where you address um know different emotional states or different um psychological States and try to personify those and try to kind of give them um uh you know abstract structure in your mind by engaging with them you know through journaling or through uh dialogue with a therapist um so that's internal family systems there's also expressive writing similarly and so if we think about um one of the impacts that um narrative can have that that that's one and sometimes it's good to um you know not sometimes it's good or sometimes it's just um a fact that we don't have narratives like we don't have narratives on all of our feelings every day I don't think we would be able to narrate everything that's that's happening to us um if we think about meditation and mantras the ideas there is that you know for a time at least we shouldn't narrate everything or at least we shouldn't accept the first kind of order narration um which we think

about mindful minut medit and kind of focusing on their breath as opposed to following our thoughts um you know if for anyone who practice meditation you know you'll know that um sometimes we should just you know be quiet and not listen um to the narratives that that show up in our brain have a couple different obviously examples um right so here what I'm talking about is and as I mentioned on the previous slide or I had it written down I don't know if I um spoke to it um but we're talking about here is that the problem of in promise of cultural narratives this is most known as Master narrative or grand narratives um that traditionally would would have been the purview of Mythology and religion on more modern times um is kind of constructed um in other ways and so um as I mentioned from our nomadic multiband past um to even agrarian societ um and then kind of coming into the universal age these um started from these fully integrated narratives where a particular nomadic um HRA tribe would have a single kind of Master cultural narrative um or at least that's what we understand as our kind of best guess is that this was the case or wasn't um you know which religion do my part of I'm part of my tribe's religion um you know why did uh we catch food today and and and not yesterday there was you know an answer from the shaman for that you this this it was fully integrated um the second kind of epoch was the leading up to kind of the the axial age when the universalizing religion you know the stage for Christianity um you know Hinduism um you know Daoism that that I mentioned um so for a very long time um this is kind of following from you know 10,000 years ago when we um started farming um perhaps up until recently we had these dominant religions um that essentially kind of took over the the sphere um those are the the universal narratives religious Nares I'm sure folks are aware of and most recently in um weird Societies or Western societies is kind of Western educated industrialized rich and Democratic um that's the acronym for for weird there um and my case here is that these are you know we're largely in a culturally fragmented environment so earlier when I mentioned single equilibrium analysis neoc classical economics but specifically um deontology um and utilitarianism those are all examples of from I think a thermodynamic systems point of view from an active inference point of view this would be kind of a um a very um single equilibrium um you know kind of ways of looking at the world um and why are they well um you know according to Joseph Enrique and I think According to some similar analysis from um Ian M christs um this is because Western people are distinctly individualistic um they're a bit more left brain um centered than um at least the axial age portion of um Society so those um traditionally of settled societies um that are more just salt and more red brain oriented um would think of things a bit more holistically uh rather than as kind of

reductive so the good of course there's um you know many benefits for left brain thinking it's um more individual it's more analytical uh and it'll be a bit more geared towards science um versus mythology versus um stories and narrative um but what is it missing well it can make a lot of um mistakes and not kind of random mistakes um a specific type of uh mistake which is um leads to kind of reductionism um and which is how we kind of ended up with which might be according to these views how we kind of ended up with um some of the paradigms that we did yeah and so the the fragmentation we can understand along kind of three different these three different dimensions right so one is aesthetic uh the next is kind of institutional and the final the last is uh kind of objective and so um this is essentially how I see our Master narrative currently broken down um and of course there could be additional ways but here's one of three ways as a starting point point to understand the kind of cultural narrative um and how specifically it's it's fragmented so aesthetic we have the kind of modern era the postmodern era and um now what some people are calling the better modern era it's maybe a bit too soon to um turn on it however I do think it's super interesting especially in the light of the free engine principle um which I'll hopefully have time at the end to kind of um touch on um the second one is our institutional um so uh the economic framework um you know how we should distribute wealth um you know what does that say about us and kind of what's the most quote unquote efficient way um but what are the narratives around um our economics what's our narratives around our politics um and finally there there's this idea of uh of you know objectivity still kind of creeping in so um what are the kind quotequote true narratives out there um and so you know a reasonable question is okay this is you know we all have kind of our own stories um perhaps there's still vague senses of a master narrative and I think you know there're always um are uh there always kind of might be um at least some emergent um Master narratives by virtue of us kind of all occupying the same at least you know digital space the same kind of time um will all Converge on some things um however without a um kind of coherent or a more of like a religious story that we um perspective that you know we had in the past where um you the narrative is not just um okay we're this you know Third Rock From the Sun we're revolving around um we wol around space and you know we're a victim to to laws and um this is what happens to be happening with Society um what we lose with the kind of grand Cosmic narratives of the past um the type of psychological um coherence um so where do we fit into this world you know the idea of going back to kind of cosmic ban you know where is our place in the universe um what is that kind of personal meaning for me that's where a um a tighter narrative kind of more intentional Master narrative

um comes into play so our current fragmentation um leads to personal anxiety it can lead to social anxiety um and it can lead to kind of the polarization uh and partisanship and the results are fragmentation institutional misalignment so whether the company you're working for um shares your value use um whether they're working on something that is you find deeply meaningful that is deeply meaningful for your identity um and your values um that's um what you mean thereby misalignment and so in terms of Aesthetics um as I mentioned um the per is is sometimes called metamodernism um and so that kind of gets into the idea of genres meaning that we have kind of plenty of good um you know television should be have plenty of good narratives available um for us to consume um that does include um kind of classic genres um but the problem is that they're all genres so we're all kind of consuming different um content we're all consuming it through different media um even in times past um not even too far past uh we would watch TV at the same time you watch the same movies because because they're in the movie theater that's where you could that's the only place you could watch them um up until recently um now we're watching movies at different times moving from movies from um 20 30 years back um from 50 years back and the essentially the fragmentation um has gone up an explanation as I mentioned there there's an analytical um orientation um and so here um I think it's it's worth noting that along with the fragmentation that has occurred in weird societies um along kind of side that that kind of loneliness that happens um with the rise of uh the analytical world is some great some great literature um and so one of the reasons I argue that this came about was specifically because um once theology once religion stopped serving the role that it did um that left in particular artists in particular those kind of aesthetically inclined to seek out um art and to create art in some cases to fill those gaps for um offering a narrative for answering the questions the big questions that were typically answered by those institutions um and so kind of going back to my quest what we're kind of after here is we're looking for greater um cultural cohesion we're aiming for better personal and social integration so I argue that these pieces are connected um and I'd argue that if the free if the free principle has told us one thing um it that agents minimize uncertainty however if it tells us another thing is that one way to verify kind of any Claim about um the self-organizing system it's to not look at one Paradigm assignments to look at it's to synthesize various different views so what we can glean from that is essentially a meta modern approach um which is to not necessarily ensure that everyone in culture everyone in a particular society is consuming the same content or media but it's to highlight certain aspects of it that each person can find meaning and the right type of meaning in their own content and even if we

achieve that through different areas will all be experiencing something similar with each of our Traditions or and once we've identified the certain components or make the case for certain components um that answer maybe some important questions from active inference point of view and allow greater visibility into that uncertainty minimizing process at a cultural scale um so here I think poetry is a great example because of it um you know it reaches everything from the senses from its sounds to its visuals um to its know deal deeper symbolic meaning and I think know great great example of that is TS Elliott um the great many references he makes in his work um and also um the kind of poetic conventions he adheres to makes it just become very uh in some cases accessible um works that he offers and so whether it's TS Elliot or Kendrick Lamar um in each case what these artists constructing is a ex symbolic self and explicated um a number of different nodes in a hierarchy um that are connected with all these various references in a particular song um that are that can be made clear you know for instance in uh um this it's genius so you can see where the different um lines in this case Kendrick Lamar has and what the references were and so in a sense what you experience when we read poetry or we read a work of fiction that has um deep references deep um cultural references like this it's constructing an extensive self um bringing coherence to it and to everyone who listens to it um and by kind of calling it out you're able to share in it almost as if you know it's something you know it's a greater work of literature in the past um as folks would and so that I think is of a few examples of um ways we can start to apply a rigorous framework to the humanities to tells what kind of literature does for us why it does what it does for us um and and parse through some of the components that make it meaningful to us and kind of allow us to um use literature um in ways that I think it was originally intended to in past times has so leave it there and um you know once again feel free to um reach out to me with my contact information and um and please you know follow my um you know substack and and let me know if um anyone has any feedback and I just want to thank Daniel Freedman again um and you know all the folks at active inference Institute and Carl priston for um the this content and um the discussions here so uh thanks again e